COMP232 Lab 9

Linux Firewall Exploration

For the purpose of this Lab session we will continue using *SEED Labs*. Please use the following references.

References:

- [SEED Project]
- [SEED Labs Firewall Exploration]
- [SEED Labs 16.04 Firewalls.pdf]
- [SEED Labs 20.04 Firewalls.pdf]

Recall that we have already have a Virtual Machine running Ubuntu 20.04 operating system with all required packages for Seed Labs in the University Labs as an application. Please locate this (on the Desktop), and run the application as you did in Lab 7.

Task 1

Revise, and expand, on the theory behind Firewalls from the lectures. In particular, make sure you understand the following, and where they operate with respect to the OSI layers:

- Packet level gateway (Layer 3)
- Firewall with *stateful packet inspection* (Layers 3,4)
- Circuit level gateway (Layer 4,5)
- Application layer gateway (Layer 7)

In addition to the lectures, use the following references to get a deeper understanding of firewalls.

References

- [OSI model - Wikipedia]
- [Packet filtering]
- [Stateful firewall]
- [Circuit-level gateway]
- [Application firewall - Wikipedia]

Task 2

- Read the Linux Firewall Exploration Lab overview SEED Project.
- Work through Task 1 (Section 2) in this link: SEED Labs 16.04 Firewalls.pdf. In particular, perform the following part (third item at the top of page 2):
  Prevent A from visiting an external web site. You can choose any web site that you like to block, but keep in mind, some web servers have multiple IP addresses.

Solving the above will require knowledge of `iptables`. Please read through Section 4.1 and 4.2 (page 8) in the new version of the SEED Labs lab sheet: SEED Labs 20.04 Firewalls.pdf

Please note that the application installed in the University Lab supports **single VM only**. So, please ignore all tasks which require two or more VMs.